

# DATA PROFILING LOG — Credit Risk

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## Data:

Total Rows: 32,852

Total Columns: 12

Columns with nulls:

- Nulls in loan\_int\_rate: 3,115 rows
- Nulls in person\_emp\_length: 894 rows

Action: Remove all null rows in Power Query

loan\_status values: 0 – No default and 1- Default

What we are building: A 3-tool portfolio project simulating a bank's retail loan portfolio risk monitoring system.

The story we're telling: "A retail bank has 28,638 loan accounts. A job as the analyst is to clean the data, score each customer's risk level, identify the high-risk segment, and present a dashboard that helps the risk committee make decisions."

This is a real Monday morning task at companies like JP Morgan, Citi, HSBC.

## Observations:

Observation 1:

- High Risk and Medium Risk tiers dominate the portfolio.
- Low Risk accounts appear to be the minority.

Hypothesis: This dataset represents a subprime lending portfolio — customers who were approved despite elevated risk profiles. This will likely show a higher than average default rate.

Observation 2:

- 22% default rate against industry average of 2-5%.
- Portfolio is significantly distressed.
- Average interest rate of 11.04% confirms subprime positioning.
- Immediate risk review recommended.

Observation 3:

- Medium Risk dominates at 45% — this is borderline acceptable.
- However High Risk + Critical Risk combined = 28.81% of portfolio.
- Nearly 1 in 3 customers is High or Critical Risk.
- Combined with 22% default rate this suggests poor credit underwriting standards at loan origination.

#### Observation 4:

- Top 3 loan purposes are Education, Medical and Venture.
  - These are non-collateralised, necessity-driven loans.
  - Customers borrowing for education and medical expenses are likely under financial stress already — higher default probability than home improvement or debt consolidation.
  - Venture loans indicate risk-seeking borrower behaviour. Combined with 22% default rate this portfolio requires immediate segmentation by loan purpose vs default rate.
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- Total Customers: 28,632
  - Portfolio Value: £276 Million
  - Avg Interest Rate: 11.04%
  - Default Rate: 22%
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- Risk Distribution:
  - Medium Risk: 45.25%
  - High Risk: 27.35%
  - Low Risk: 25.94%
  - Critical Risk: 1.46%
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- Defaults: 6,000 customers
  - Non-Defaults: 22,000 customers
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- Top Loan Purposes:
  - 1. Education- 5,700
  - 2. Medical - 5,300
  - 3. Venture- 5,000
  - 4. Personal/Debt Consolidation
  - 5. Home Improvement

#### Data Consistency Note:

- SQL row count: 28,638
- Power BI row count: 28,632
- Difference: 6 rows

Reason: Power Query age filter (>80) was applied in Power BI but not replicated in SQL import.

SQL analysis includes 6 additional records with age above 80.

For precise consistency add WHERE person\_age <= 80 to all SQL queries.

#### Observation 5:

- 55% of customers fall in medium income band (30k-70k).
- Only 8.7% are Very High-income earners.

- Combined with 22% default rate this suggests the portfolio is heavily exposed to middle income borrowers who are financially stretched — most vulnerable to interest rate rises and economic downturns.

Data correction: Income Band formula corrected during build — Medium threshold fixed from 7,000 to 70,000.

#### Observation 6 — KEY FINDING:

- Risk Tier classification validates as a strong default predictor.
- Critical Risk customers default at 69% — nearly 7 in 10.
- High Risk customers default at 40% — nearly 1 in 2.
- Low Risk customers default at only 9% — well controlled.
- Tightening credit standards for interest rates above 13% could reduce portfolio default rate significantly.

Recommendation: Bank should immediately review lending criteria for High and Critical Risk segments.

#### Observation 7:

- Debt Consolidation: 28% default rate
- Medical: 27% default rate
- Home Improvement: 26% default rate

Counterintuitive finding — Debt Consolidation has highest default rate at 28%, not Education or Venture as initially hypothesised.

Interpretation: Customers consolidating debt are already financially distressed at loan origination. They are borrowing to pay off existing debt — a sign of cascading financial stress.

These customers have highest default risk despite appearing to be responsible borrowers.

Recommendation: Debt Consolidation applications should trigger enhanced credit scrutiny and lower loan-to-income ratios at approval stage.

#### Observation 8:

- Employment stability shows expected inverse relationship with default rate for 0-15 years.
- Unexpected spike at 25+ years suggests mid-to-late career financial stress — possibly retirement planning pressure, dependent family expenses, or accumulated debt burden.

Statistical note: Data points beyond 35 years employment have small sample sizes — findings not statistically reliable at extreme ends.

Recommendation: Employment years should be used as a secondary risk indicator, not primary. Most predictive in the 0-5 year range.

Observation 9:

Perfect progressive relationship between Risk Tier and average loan amount. Critical Risk customers borrow 70% more than Low Risk (£14,600 vs £8,600).

Combined with 69% default rate for Critical Risk:

These customers are borrowing the most AND repaying the least.

This represents the highest loss exposure in the portfolio.

Recommendation: Implement loan amount caps for High and Critical Risk customers. Maximum loan amount should be calibrated against income and credit history, not just credit score.

Observation 10:

Credit history length shows negligible predictive power for default risk in this dataset. Flat trend from 0-20 years suggests length alone is insufficient as a risk indicator.

Limitation: Dataset lacks credit score quality metrics which would normally accompany history length in a real banking system. Slight uptick after 20 years consistent with Observation 8 — age-related financial stress pattern emerging across multiple variables.

Recommendation: Credit history length should be deprioritised as a standalone risk indicator. Replace with credit score or payment delinquency count for stronger predictive power.

Observation 11 —

**CRITICAL FINDING:**

- Loan Grade shows perfect progressive default relationship from Grade A (lowest) to Grade G (highest).
- This cross-validates our Interest Rate based Risk Tier classification from Page 2.
- Two independent risk models agree on the same conclusion — increases confidence in our risk assessment framework.
- In a real banking environment this alignment would justify using Interest Rate as a viable proxy for internal loan grade when grade data is unavailable.

Observation 12 —

**MOST IMPORTANT BUSINESS FINDING:**

- Counterintuitive risk concentration discovered. Critical Risk has highest default RATE at 69% but only £6M total exposure —manageable loss scenario.
- Medium Risk has only 16% default rate BUT £126M exposure — largest systemic risk in the portfolio.
- A 10% deterioration in Medium Risk repayment behaviour would cost the bank £12.6M — 3x the entire Critical Risk portfolio value.

Recommendation: Immediate portfolio health monitoring program for Medium Risk segment. Early intervention and payment support schemes should be prioritised for this segment over Critical Risk due to scale of financial exposure.

#### Observation 13 — ACTIONABLE POLICY FINDING:

Clear default rate inflection point identified at 12% interest rate threshold.

Below 12%: default rate 9-15% — within manageable range

Above 12%: default rate jumps to 31-86% — unsustainable

#### Recommended Credit Policy Change:

Loans above 12% interest rate should require:

1. Enhanced credit committee approval
2. Lower loan-to-income ratio limits
3. Mandatory collateral assessment
4. Quarterly portfolio review trigger

This single policy change could reduce portfolio default rate from 22% to approximately 12% by restricting the highest risk approvals.

#### Observation 14:

- Portfolio heavily concentrated in young borrowers peaking at age 23.
- Young borrowers + Education/Medical loan purposes +
- Low financial buffers = explains elevated default rate.

Recommendation: Age-adjusted credit scoring should be implemented. Young borrowers need income trajectory assessment not just current income snapshot.

#### Observation 15:

- 92% of borrowers either rent or have a mortgage — meaning virtually no customers have unencumbered assets.
- This portfolio has extremely low collateral coverage.
- In a default scenario the bank has almost no assets to recover losses against.

Recommendation: Collateral assessment should be mandatory for all High and Critical Risk applications.

Secured lending products should be developed for mortgage and rent paying customers to reduce unsecured exposure.

Observation 16 —

**COMPLETE BORROWER PROFILE:**

- Dominant customer segment is young urban renters aged 20-25 with under 3 years employment history borrowing for education and medical necessities.
- This segment combines every major default risk factor: youth, employment instability, no collateral, necessity-driven borrowing.
- Portfolio was likely built targeting underserved young professionals — high social value but requires specialist risk management approach.